

Exercise 2. Existence of graphs with given degree sequences

Solution. We use the **Erdős–Gallai theorem**: a non-increasing sequence $d_1 \geq \dots \geq d_n$ of non-negative integers is graphical if and only if

$$\sum_{i=1}^n d_i \text{ is even, and for each } k = 1, \dots, n: \quad \sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k).$$

We also use the elementary bound $d_i \leq n-1$.

a) $(6, 2, 2, 2, 1, 1)$: $n = 6$, but $d_1 = 6 > n-1 = 5$. **No such graph exists.**

b) $(5, 3, 3, 3, 3, 1)$: sum = 18 (even), $n = 6$, $d_1 = 5 = n-1 \checkmark$.

k	condition checked
1	$5 \leq 0 + (1+1+1+1+1) = 5 \checkmark$
2	$8 \leq 2 + (2+2+2+1) = 9 \checkmark$
3	$11 \leq 6 + (3+3+1) = 13 \checkmark$
4	$14 \leq 12 + (3+1) = 16 \checkmark$
5	$17 \leq 20 + 1 = 21 \checkmark$

Such a graph exists.

c) $(5, 4, 4, 3, 3, 2)$: sum = 21 (odd). **No such graph exists.**

d) $(5, 5, 5, 5, 3, 3)$: sum = 26 (even), $n = 6$, $d_1 = 5 = n-1 \checkmark$. At $k = 4$: $5+5+5+5 = 20$ vs. $12 + \min(3, 4) + \min(3, 4) = 12 + 3 + 3 = 18$. Since $20 > 18$, **no such graph exists.**

e) $(5, 5, 4, 3, 3, 2)$: sum = 22 (even), $n = 6$, $d_1 = 5 = n-1 \checkmark$.

k	condition checked
1	$5 \leq 0 + (1+1+1+1+1) = 5 \checkmark$
2	$10 \leq 2 + (2+2+2+2) = 10 \checkmark$
3	$14 \leq 6 + (3+3+2) = 14 \checkmark$
4	$17 \leq 12 + (3+2) = 17 \checkmark$
5	$19 \leq 20 + 2 = 22 \checkmark$

Such a graph exists.

f) $(5, 5, 3, 3, 2, 2)$: sum = 20 (even), $n = 6$, $d_1 = 5 = n-1 \checkmark$.

k	condition checked
1	$5 \leq 0 + (1+1+1+1+1) = 5 \checkmark$
2	$10 \leq 2 + (2+2+2+2) = 10 \checkmark$
3	$13 \leq 6 + (3+2+2) = 13 \checkmark$
4	$16 \leq 12 + (2+2) = 16 \checkmark$
5	$18 \leq 20 + 2 = 22 \checkmark$

Such a graph exists.

g) $(7, 6, 5, 4, 3, 3, 2)$: $n = 7$, but $d_1 = 7 > n-1 = 6$. **No such graph exists.**